

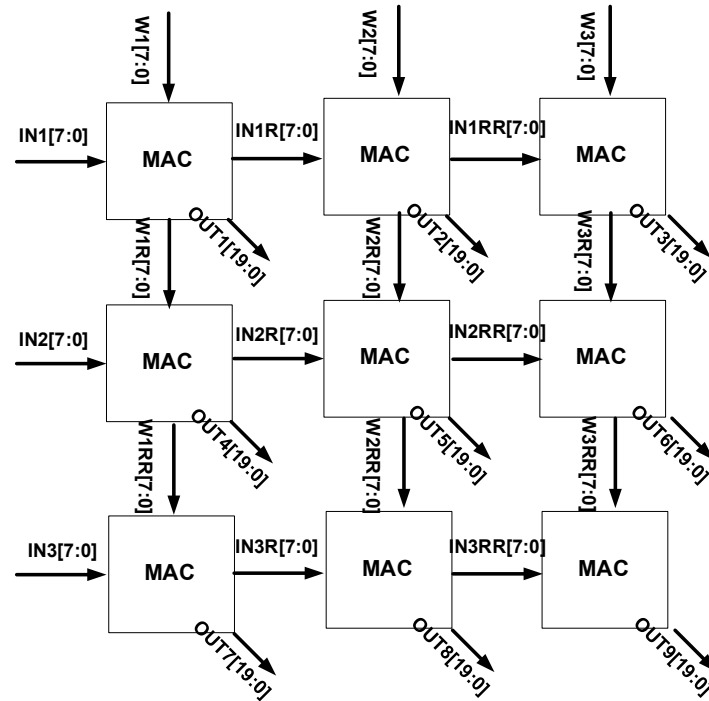
Lab #3

EECS 303: Advanced Digital Logic Design

(Due by 12:00pm 6/13/26)

1. **Background:** On-chip neural network accelerator is used to accelerate the modern AI operation on a chip. It typically consists of a large array of 2-dimensional multiplication-accumulation (MAC) modules. For instance, Google's Tensor-Processing-Unit (TPU) chip consists of a MAC array of 256x256. In this lab, you will design a mini-MAC array with a pipelined data and weight flow, called "systolic array", which is one of the most popular architecture for on-chip neural network. Each MAC unit is also referred as a processing element (PE). So the design is also referred as a PE array.
2. **Description:**
 - a. Inputs: stream of IN1, IN2, IN3 (each of signed 8 bits), stream of W1, W2, W3 (each of signed 8 bits as weights for neural network), clk (clock), rstb (resetb, reset at 0). At each clock cycle, each IN and W will have a new input arrived at the pins of the design.
 - b. Output: OUT (signed 20 bits) for each MAC unit; There are a total of 3x3 MAC units in your design;
 - c. At every clock cycle, each MAC will perform $IN \cdot W$ and accumulate with its internal stored value to generate the OUT value. At the clock cycle of 13, the MAC output will not change any more. The OUT value will remain constant until reset from rstb pin again.
 - d. The diagram of the systolic MAC array is shown in the figure below. INR1, INR2, INR3 are the same values as IN1, IN2, IN3 except that they are pipelined from IN1, IN2, IN3 (meaning they pass through flip-flops so that they are one clock cycle delayed from IN1, IN2, IN3). Similarly, INRR1, INRR2, INRR3 are one clock cycle delayed from INR1, INR2, INR3. The same operation applies to W1, W2, W3. W1R, W2R, W3R are one clock cycle delayed from W1, W2, W3. W1RR, W2RR, W3RR are one clock cycle delayed from W1R, W2R, W3R.
 - e. The MAC design runs at 0.5GHz clock;
 - f. All inputs, i.e. IN1, IN2, IN3 and weights W1, W2, W3 (except clk, resetb) need to satisfy a max input delay of 0.5ns and a minimum input delay of -0.2ns from external modules;
 - g. All output needs to satisfy a max output delay of 0.5ns and a minimum output delay of -0.2ns (The negative minimum delay is used to add additional hold margin for the output signal paths);
 - h. At beginning, a reset is issued. Upon reset (rstb =0), all internal flip-flops are reset to 0;

- i. Place the IN1, IN2, IN3 pins at the left boundary of the design; Place W1, W2, W3 pins at the top boundary of the design; Place OUT1 to OUT9 pins at the bottom boundary of the design.



3. Deliverable:

- (1). Following the lectures, create all required design files including RTL of MAC array, testbench, timing constraint SDC file, etc.
- (2). Use Xcelium to simulate your RTL using your testbench. Show the snapshots of the simulated waveforms (show all inputs and outputs signals) for fifteen continuous clock cycles. You can split into multiple pictures if the pictures are too crowded to be seen. The inputs should change every clock cycle (It is your choices of what IN and W to give, but cannot be zeros). Show waveforms of your simulation and highlight (circle) all the correct outputs values at the end. Show every output is calculated correctly by giving a table of anticipated value and measured value from the waveform).
- (3). Perform synthesis using Genus.
- (4). Perform backend design using Innovus.
- (5). Perform gatelevel simulation (either delay annotated or zero delay mode is fine). Show waveforms of your simulation and highlight (circle) all the correct outputs values. Show every output is calculated correctly by giving a table of anticipated value and measured value from the waveform).
- (6). Submit a single report file in word or PDF containing the following:
 - a. waveforms and tables from (2) and (5)
 - b. Snapshot of your finished backend layout of your design; Show the X and Y dimension of your design;
 - c. Your RTL and testbench codes;

- d. Your SDC file;
- e. The timing report from your finished backend design for both setup and hold analysis. Only report on the worst-case timing path (both setup and hold) is needed. The timing report can be found in “timingReports” folder;
- f. The area report from your finished backend design;
- g. verify_drc result from your finished backend design;
- h. verify_Connectivity result from your finished backend design;
- i. The power analysis result (use Input Activity of 0.2 and Dominant Frequency of 0.5GHz) from your finished backend design;
- j. Please also include the path of your design folder in Wilkinson’lab.